Superstore Sales & Profit Optimization Analysis

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Tools : Python & Power BI

1. Project Overview

This project presents an end-to-end analysis of store sales data to uncover key business insights related to sales performance, profitability, customer behavior, and operational efficiency.

Using Python for data preprocessing and Power BI for visualization, the project transforms raw transactional data into interactive dashboards that support data-driven decision-making.

The Project highlights critical issues such as profit leakage, discount impact, and delivery inefficiencies, enabling businesses to optimize strategies and improve overall performance.

2. Dataset Description

The project uses the Superstore retail dataset, which contains transactional data related to sales, customers, products, and orders.
The dataset provides detailed information required to analyze business performance, profitability, and customer behavior across different regions and segments.

Key Columns:

- Order ID
- Order Date
- Customer ID
- Region / City
- Category / Sub-Category
- Sales
- Profit
- Discount
- Ship Mode

3. Data Cleaning & Preparation in Python

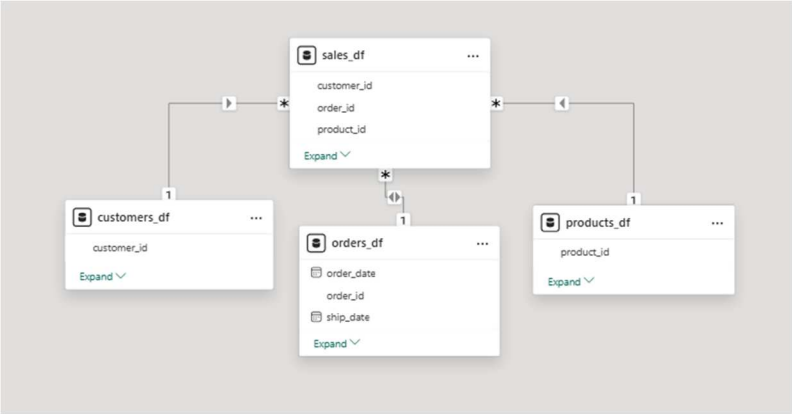
The raw dataset was cleaned and transformed using Python to ensure accuracy, consistency, and efficient analysis.
Data preprocessing was performed using libraries like Pandas and NumPy to prepare the data for modeling and visualization key steps like.

- Removed missing and null values to maintain data quality.
- Eliminated duplicate records to avoid data redundancy.
- Converted date columns into proper datetime format for time-based analysis.
- Standardized categorical values Ex. category names, regions
- Created new calculated like Profit Ratio
- Split the dataset into multiple structured tables (customers, products, sales, orders)

- And the last prepared clean and structured data for efficient use in Power BI

4. Data Modeling

The dataset was structured into a relational data model to enable efficient analysis and better performance in Power BI.



A star schema approach was used to organize the data into fact and dimension tables, ensuring scalability and optimized query execution and also make proper relationship with each dimension table.

5. Analysis, Insights & Recommendations

1. **Are sales increasing over time?**

Ans:

Yes, Sales are increasing over time. In 2014, sales were $405.57K, then it slightly decreased in 2015 to $427.12K. After that, sales started increasing again in 2016 ($549.26K) and reached the highest in 2017 ($659.80K).

This shows that even though there was a small drop in 2015, business recovered and grew strongly in the following years.

2. **Is profit growing at the same rate as sales?**

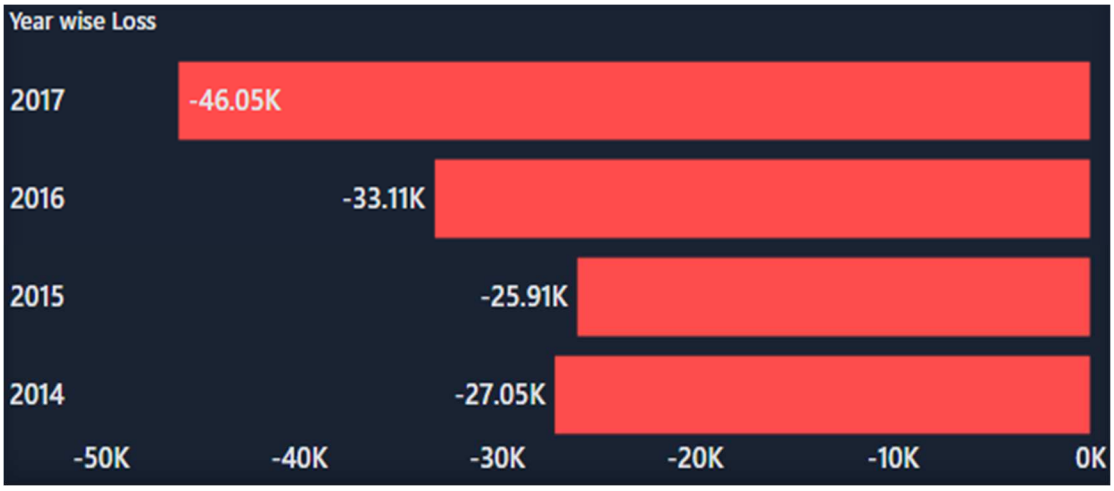
Ans:

Profit is increase over the time. In 2014, profit was $43.99K, then it increased to $60.49K in 2015, $76.44K in 2016 and reached $89.05K in 2017.

This is shows that the profit is growing every year, even sales is drop in 2015. However, we cannot clearly say that profit is growing as the same rate as sales, because both are not increasing proportionally.

3. **How is loss changing over time?**

Ans:



The total loss is increase over time. In 2014 the total loss was $27.05K, then it slightly decreased to $25.91K in 2015. However, after that it increased and reached the highest level in 2017 ($46.05K).
This shows that even though sales and profit are increasing, the loss is also increasing over time.

This indicates that the company is growing, but at the same time it is also facing

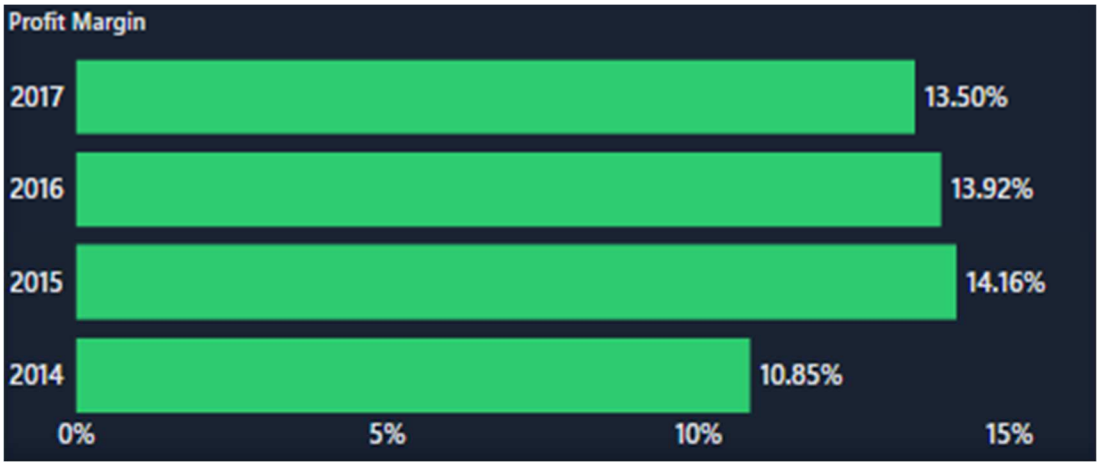
higher losses in some areas.

We should focus on:

- Price optimization.
- Reducing high discount.
- Identifying and avoiding loss making products.

4. How is profit margin changing over time?

Ans:



The profit margin increased from 10.85% in 2014 to 14.16% in 2015, but after that it slightly decreased to 13.92% in 2016 and 13.50% in 2017.

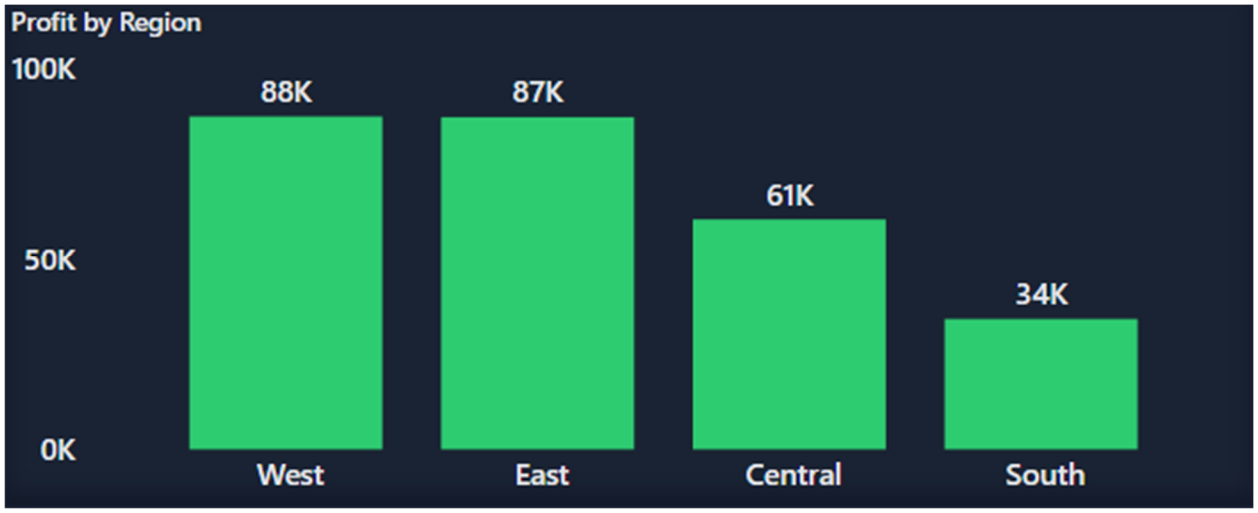
This shows that the store improved its profitability in 2015, but was not able to maintain that level in the following years.
Even though sales and profit are increasing, the decreasing profit margin after 2015 indicates that the costs or discounts might be increasing.

Suggestions:

- Control discount and pricing strategy.
- Focus on high margin products.
- Reduce unnecessary costs

5. **Which region generates the highest profit?**

Ans:



The west region generates the highest profit ($87.66K), it is very close to the east region ($87.50K). The south region has profit of $34.20K, While the central region has lowest profit of ($16.61K).

West and East region are performing very well and contributing majority of the profit, while center is significantly underperforming.

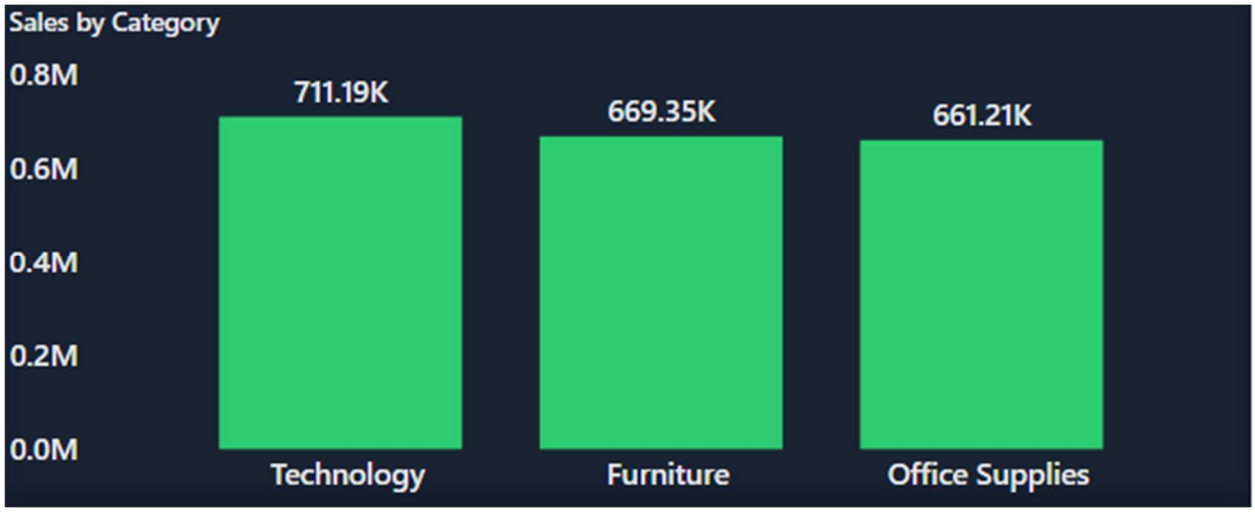
The business is highly dependent on west and east region for profitability, and central region may have issues like high discount or low pricing.

Work on:

- improve performance in central region.
- Analyze why west and east are successful and replicate their strategies in the central and south region.
- Optimizing price and Discount strategies in low performing regions.

6. **Which category contributes most to sales?**

Ans:



The technology category contributes the most to sales with $711.19K, followed by furniture at $669.35K and office supplies at $661.21K.

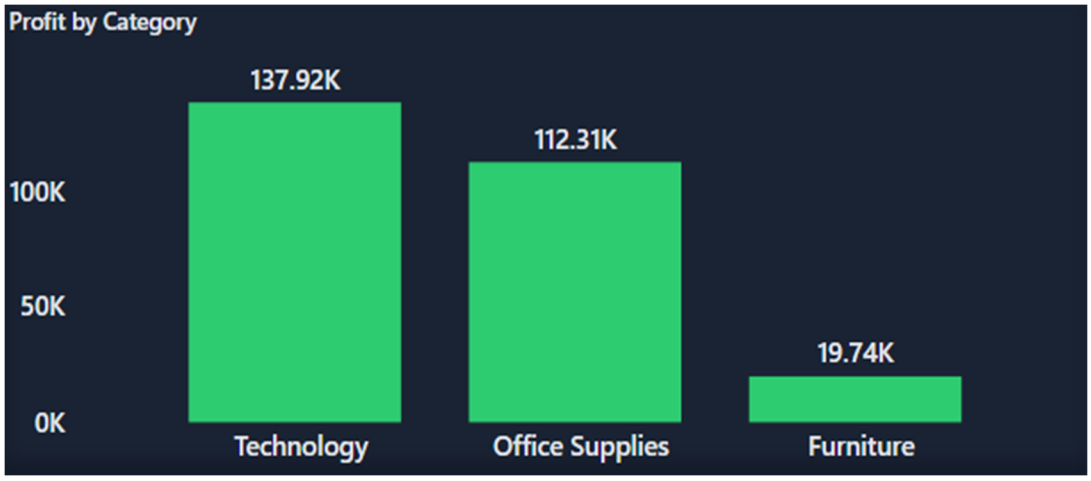
Technology has the highest sales among all categories, while furniture and office supplies have slightly lower but comparable sales,

Focus on:

- Continue focusing on Technology products to maintain high sales.
- Improve profitability in other categories like furniture.

7. **Which category contributes most to profit?**

Ans:



The technology category contributes the most to profit with $137.92K, followed by office supplies at $112.31K, while furniture has lowest profit at $19.74K.

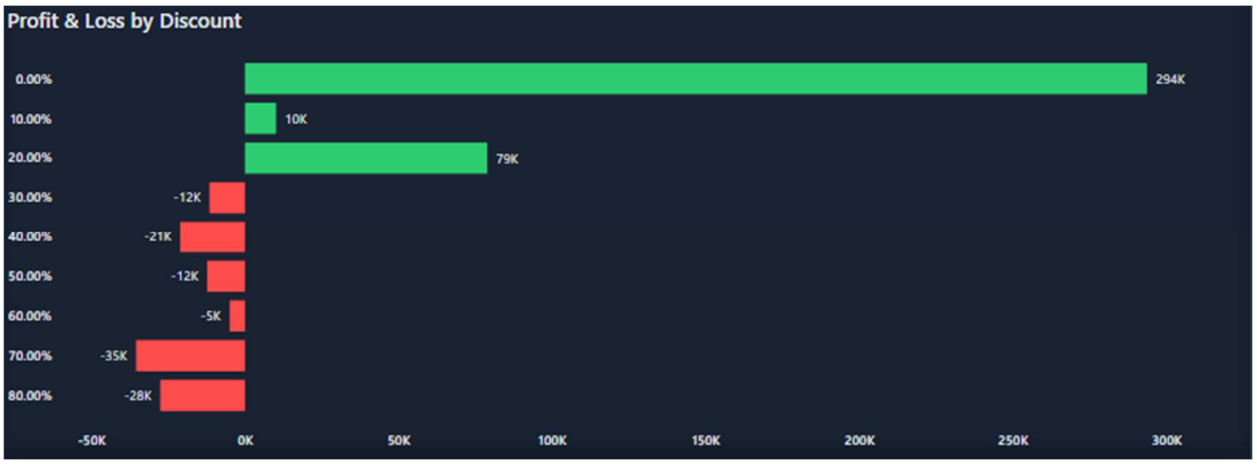
Technology generates the highest profit among all the categories, and office supplies also perform well. However, Furniture has very low profit compared to its sales.

Suggestions:

- Focus more on technology products to maintain the high profit.
- Improve pricing and reduce discounts in furniture category.
- Identify and fix loss making products in furniture.

8. How does discount impact profit?

Ans:



When the is between 0% to 20% generally the product makes profit. However, When the discount is greater than 20%, Products start making losses.

At lower discount levels, the company is able to maintain a good margin. But as the discount increases beyond 20%, the profit decreases and turns negative.

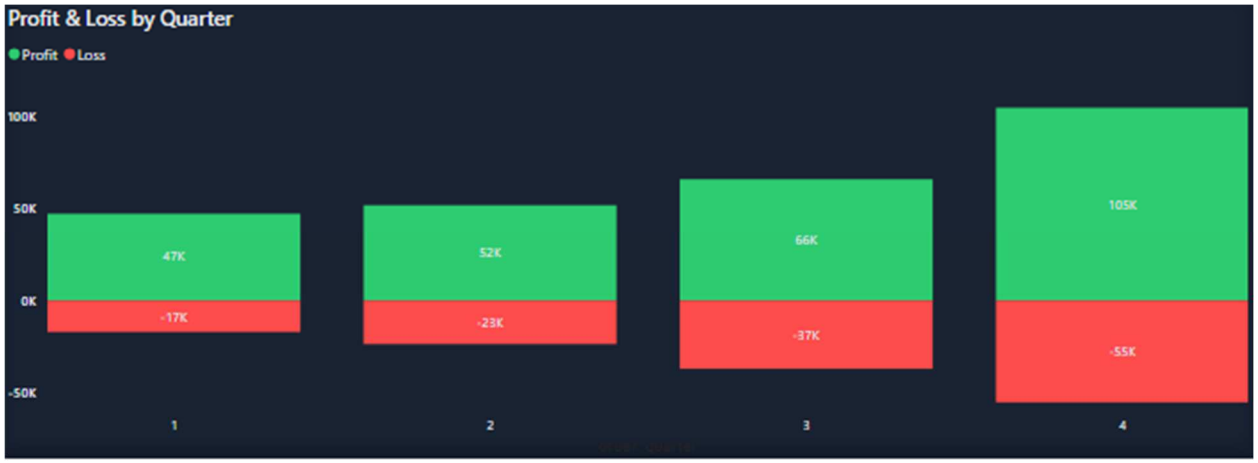
we can say that the high discount directly reducing profitability. This shows that aggressive discounting is not sustainable for the business.

Suggestions:

- For increasing the profit limit discount to maximum 15% to 20%.
- Apply higher discounts only on high margin products.
- Change the discount strategy for loss making products.

9. How does profit and loss vary quarter-wise?

Ans:



In quarter 1, the total sales are $298.61K, profit is $47.38K, and loss is $16.94K
In quarter 2, the total sales are $406.70K, profit is $51.98K, and loss is $23.45K
In quarter 3, the total sales are $553.82K, profit is $66.01K, and loss is $36.80K
In quarter 4, the total sales are $782.62K, profit is $104.60K, and loss is $54.92K

Sales increases from quarter 1 to quarter 4, and profit also increase overall. However, the loss is also increasing especially is quarter 4 it becomes very high.

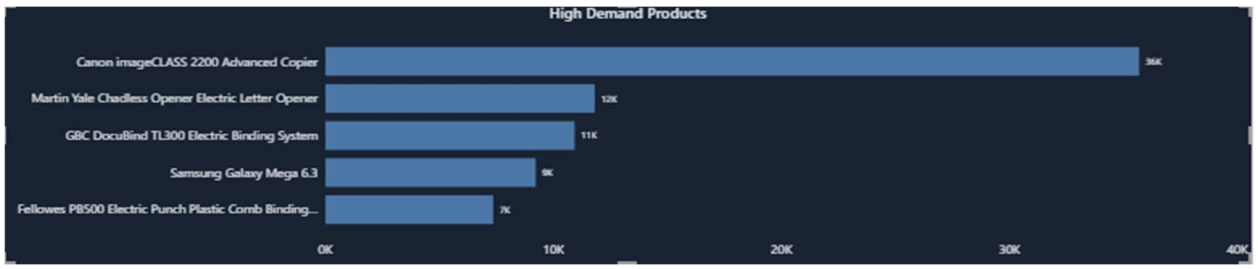
Even though quarter 4 has highest sales and profit, it also has highest loss. This means higher sales are also bringing higher losses, which may reduce overall profitability.

Suggestions:

- Stop selling loss making products.
- Control discount and pricing during high sales periods.
- Focus on reducing loss while maintaining sales growth.

10. Which products generate the highest profit?

Ans:

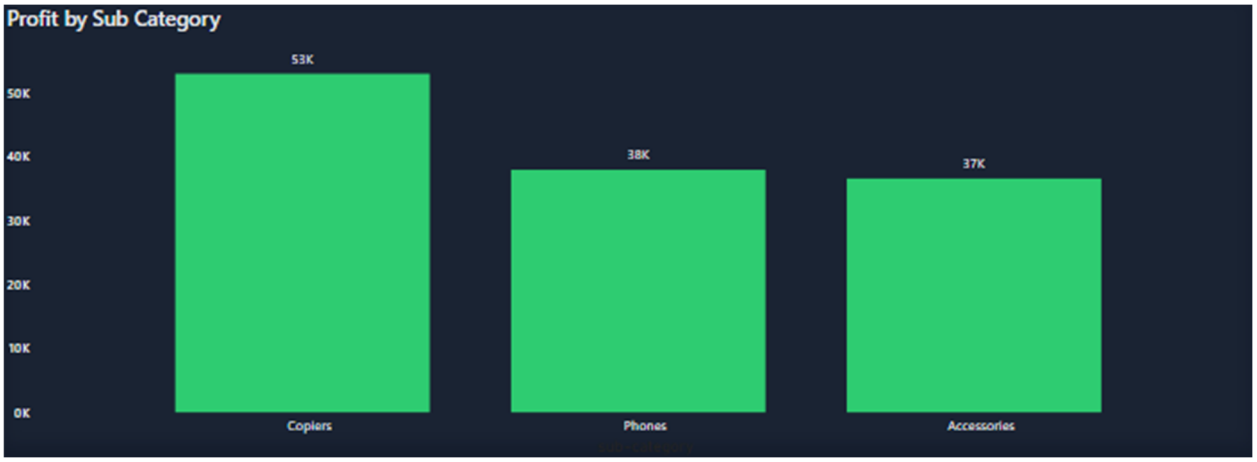


The Canon image class 2200 Advanced Copier generates the highest profit, with a profit of $15,679.96K.

This product has the highest profit among all products, which means it is preform very well compared to others.

11. Which sub-category contributes most to profit?

Ans:

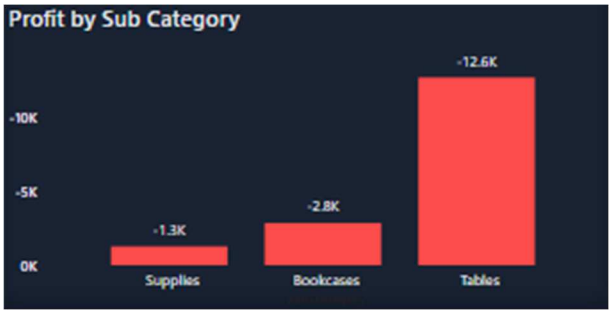


The Copiers sub category contributes most to the profit, with a total profit of $52,894.88

Copiers generates higher profit compared to other sub categories. Making it high performing segment.

12. Which sub-category is making losses?

Ans:



There are three sub categories making losses.

- Tables: -$12,644.63

- Bookcases: -$2,844.06
- Supplies: -$1,265.43

These sub categories have negative profit, which means the cost or discount is higher than the revenue generated.

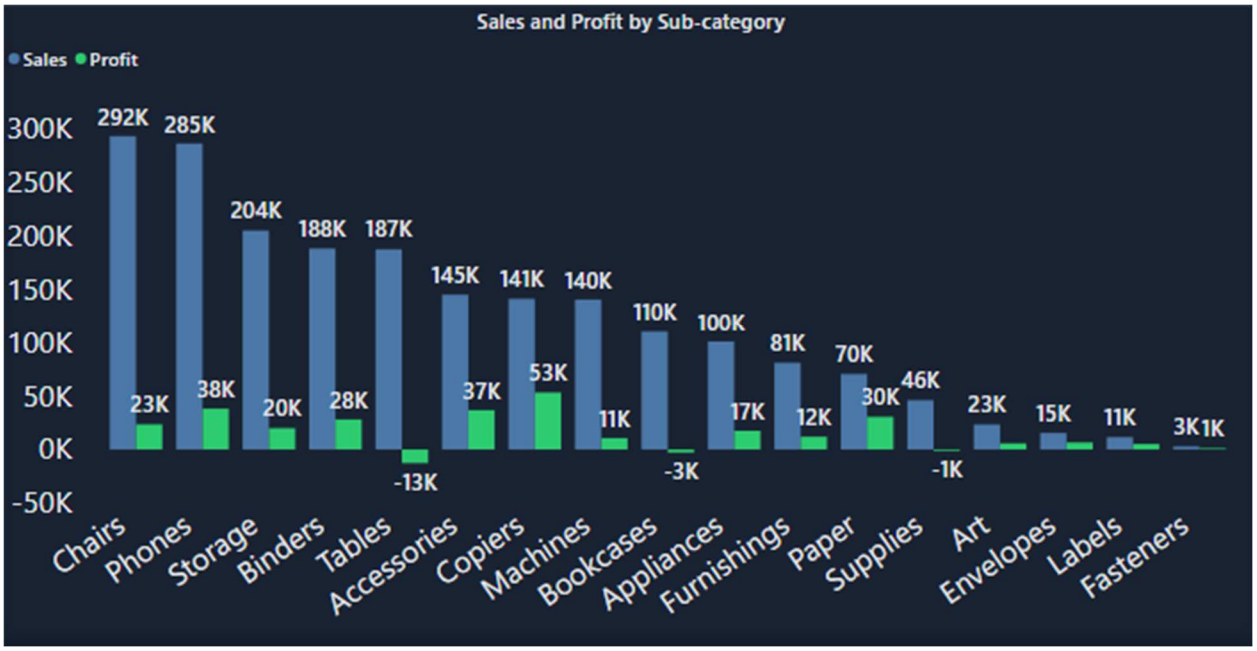
Tables are causing the highest loss, so they are main problem area. Bookcases and Supplies are also contributing to the losses but at a lower level.

Suggestions:

- Reduce discounts on Tables, Bookcases and Supplies.
- Review pricing strategy for these sub categories.
- Focus on improving profit margins.

13. Are there sub-categories with high sales but low profit?

Ans:



Yes, some sub-categories have high sales but low or negative profit. For example, Table have high sales but show a loss (-$13K). and Bookcases also have good sales but very low profit ($11K).

Form the sales and profit comparison, it is clear that certain sub-categories are able to generate strong revenue but fail to maintain profitability. The gap between sales and profit indicates that higher revenue does not always lead to better financial performance.

Sub-category like, Tables are major concern because they are not only low in profitability but are actually causes losses despite good sales. This suggests inefficiencies in pricing, cost management or discount strategies.

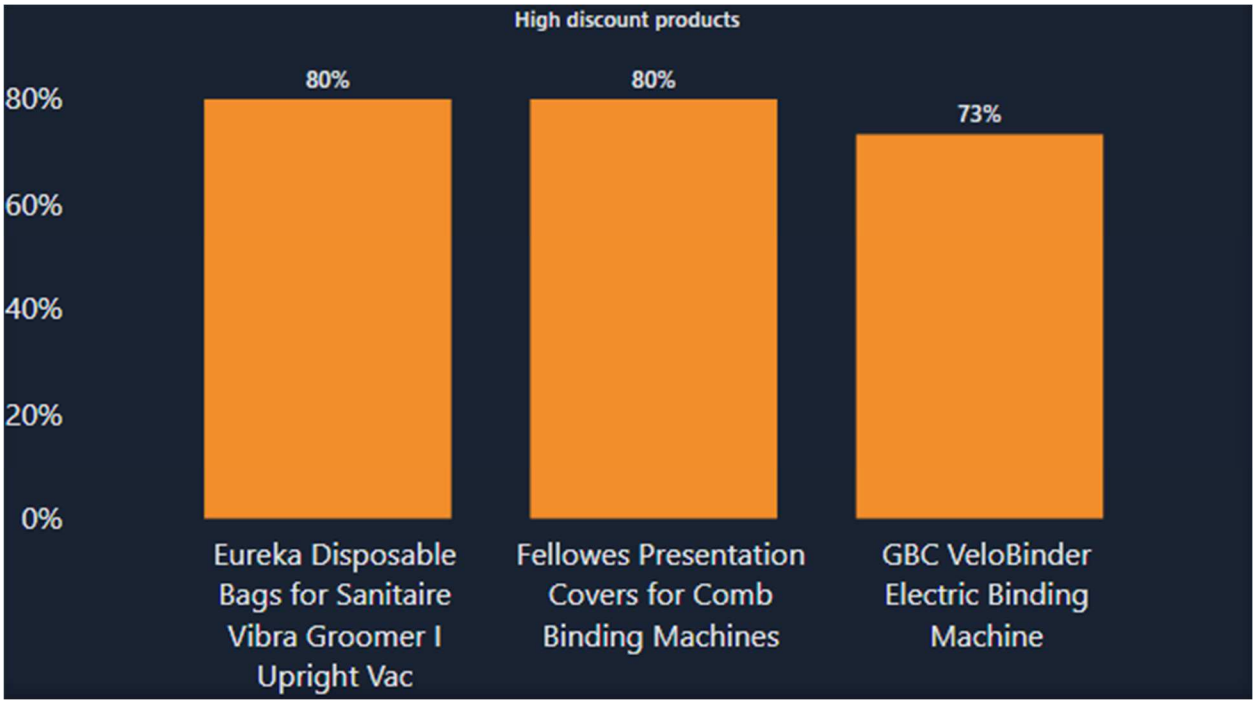
Relying on sales alone can be misleading. If high sales products are not profitable, they can negatively impact on business performance and reduce total profit.

Suggestions:

- Re-evaluate pricing strategy for loss making sub-categories.
- Control excessive discount, especially on high sales items.
- Analysis cost structure to identify areas where cost can be reduced.
- Focus on promoting sub-categories that generate high sales and high profit.

14. Which products are heavily discounted?

Ans:



The most heavily discounted top 3 products are:

- Eureka Disposable Bags for Sanitaire Vibra Groomer I Upright Vac – around 80% discount
- Fellowes Presentation Covers for Comb Binding Machines – around 80% discount
- GBC VeloBinder Electric Binding Machine – around 73% discount

The chart shows that these products are being sold with very high average discount, significantly higher than normal business levels. Discount close to or above 70% - 80% indicate aggressive pricing strategies.

Such high discount levels are likely reducing price margin and may even lead to losses, especially if the product cost is not low enough to support such discount.

Heavy discount can increase sales volume, but it negatively impacts profitability.

Suggestions:

- Reduce discount levels on these products.
- Apply high discounts only when necessary (like low demand products).
- Evaluate whether these products are profitable even after discount.
- Align discount strategy with profit margins.

15. Do high-discount products generate profit or loss?

Ans:

High discount products generally generate losses. Especially when the discount is greater than 50%, most products start making negative profit.

From the data, it is clear that as the discount increases, profit decreases. While higher discount may increase sales volume, but they significantly reduce profit margin, and often lead to loss.

High discounts are not suitable for business, even though they help in boosting sales, they fail to generate profit and can negatively impact overall performance.

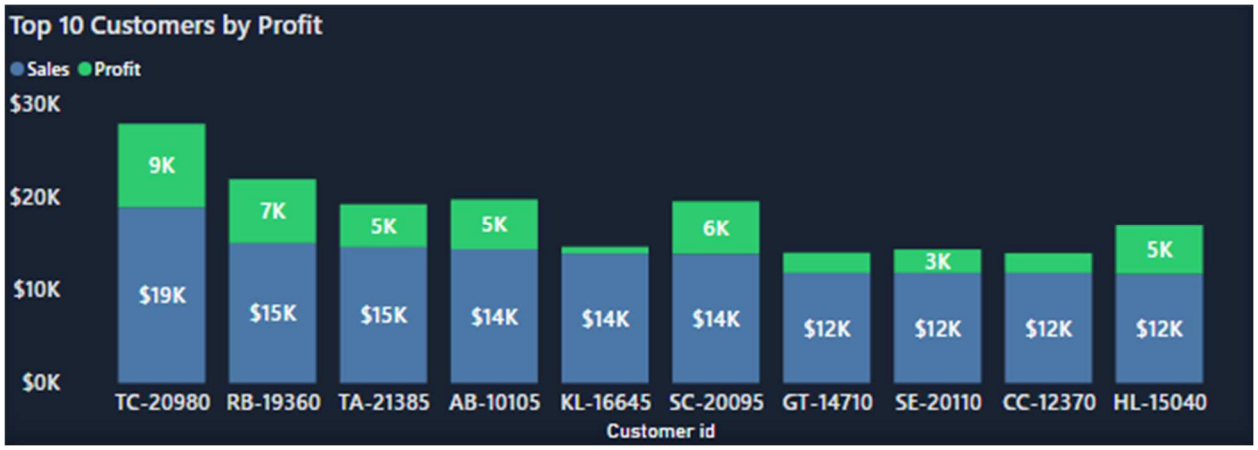
The company might be focusing too much on increasing sales through discounts, but this strategy hurting profitability.

Suggestions:

- Reduce very high discount level specially above 50%
- Improve pricing strategy to maintain a healthy profit margin.
- Avoid or limit the sales of products that consistently generate loss.
- Apply high discount only on products with high margins.

16. Which customers generate the highest sales and profit?

Ans:



The customer TC-20980 generates the highest sales $19K with the highest profit $9K. other top customers include RB-19360 with $15K sales and $7K profit, and SC-20095 with $14K sales and $6K profit.

These costumers contribute significantly to both revenue and profit. The data shows that the small group of customers play major role in overall business performance.

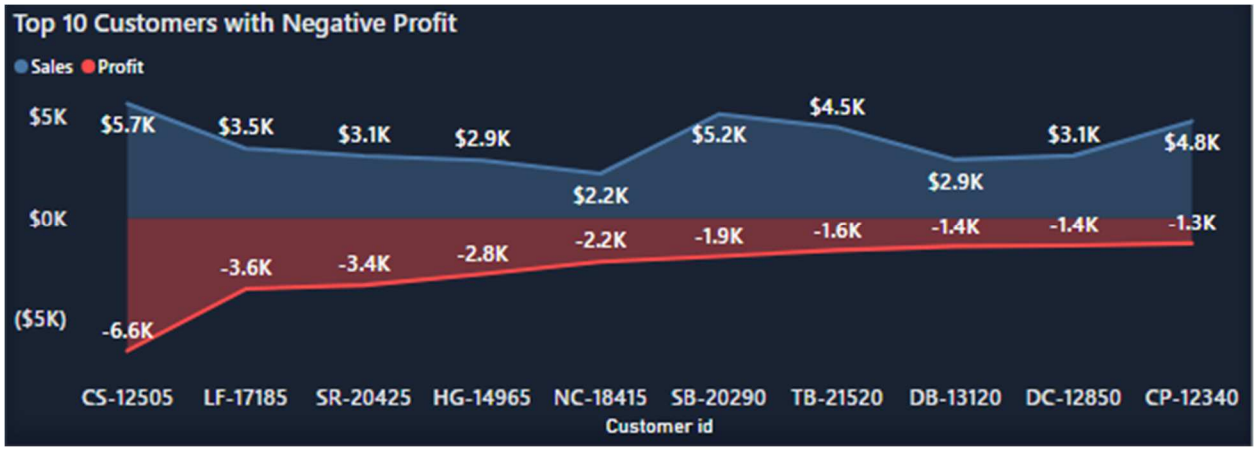
Customers like TC-20980 are highly valuable as they generate high sales and high profit making them key contributors to business growth.

Suggestions:

- Focus on retaining high value customers.
- Provide personalized offers or loyalty benefits for top customers.
- Identify and target similar customers to increase revenue and profit.

18. Which customers are causing losses?

Ans:



The customers causing the highest losses are CS-12505 (-$6.6k), LF-17185 (-$3.6k), SR-20425 (-$3.4k), HG-14965 (-$2.8k) and there are so many other customers also contributes to losses.

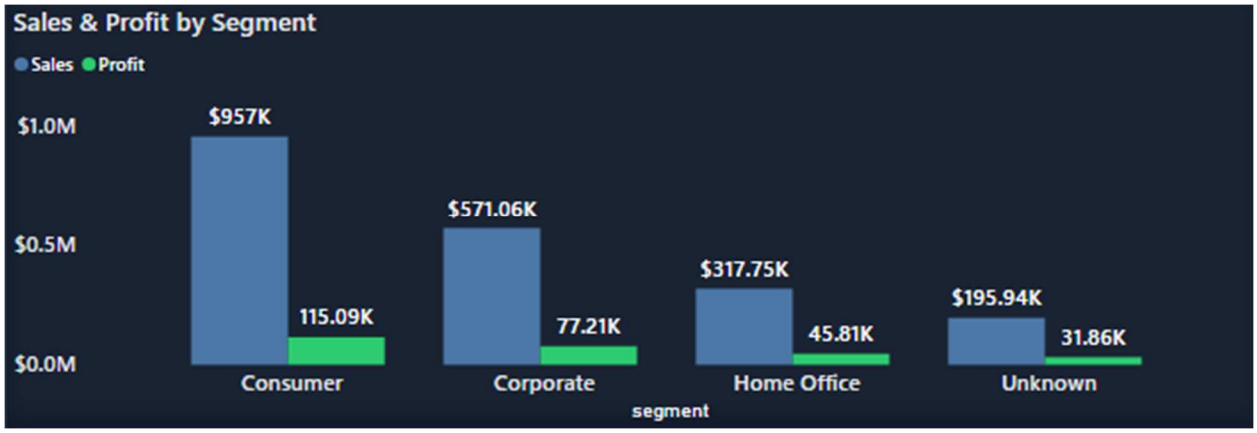
These customers consistently generate the negative profit, which means cost of serving them is higher than the revenue generated.

Suggestions:

- Analyze discount and pricing strategy for these customers.
- Avoid giving high discounts to loss making customers.
- Focus on converting these customers into profitable once.
- Limit unprofitable transactions.

19. Which customer segment contributes the most to sales and profit?

Ans:



Consumer segment contributes most to both sales and profit, with around $975K in sales and $115.09K in profit. It is followed by the corporate segment ($571.06K sales, $77.21K in profit), and home office ($317.75K sales, $45.81K profit). The unknown segment has the lowest contribution in sales and profit.

The consumer segment dominates in both revenue and profit, showing the strong contribution compared to others segments. Corporate and Home office also perform well at lowest level.

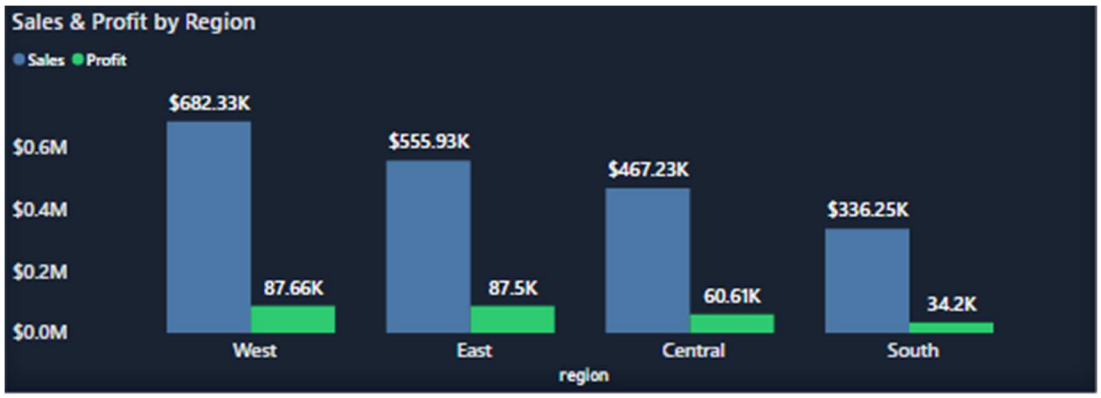
The business is highly dependent on the consumer segment for overall performance, making it the most important costumer segment.

Suggestions:

- Focus on retaining and targeting consumer customers.
- Improve price and discount strategy for corporate and home office segment.
- Increase marketing efforts towards corporate and home office segment.

20. Which region contributes the most to sales and profit?

Ans:



The west region contributes the most to both sales and profit, with approximately $682.33K in sales and $87.66K in profit. It is followed by the east region ($555.93K sales, $87.5K profit), then central ($467.23K sales, $60.61K profit), and south has lowest contribution ($3336.25K sales, $34.2K profit).

West and East are the top performing region, which contributes the highest share in both sales and profit. Central and south region lag behind in comparison.

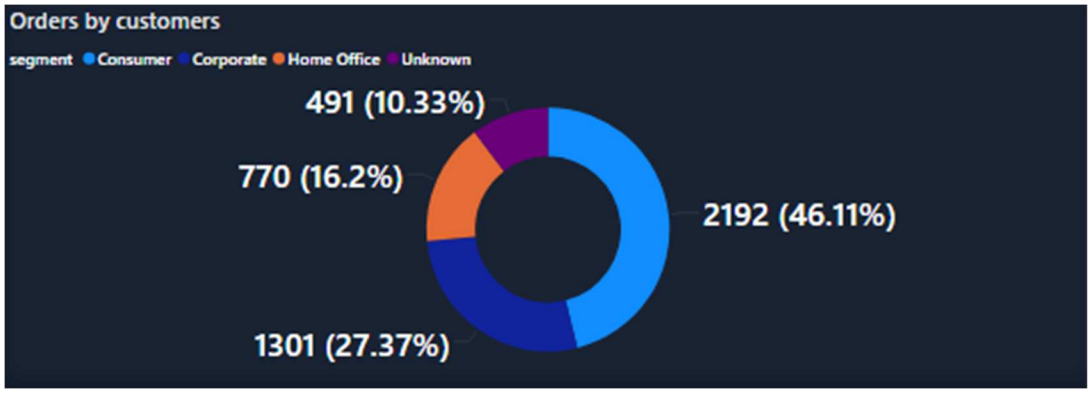
The business performance is highly driven by the west and east region, while central and south shows relatively weaker performance.

Suggestions:

- Focus on maintaining strong performance in west and east region.
- Analyze successful strategy in these regions and apply them to central and south region.
- Improve pricing, discounting and marketing strategy in central and south region.
- Identify and analyze loss making customers.

21. Which customers segment place the most orders?

Ans:



The Consumer segment place the most orders, with 2192 orders (46.11%), it followed by the Corporate with 1301 orders (27.37%), Home office with 770 orders (16.2%). The Unknown segment hast the lowest number of orders 419 with (10.33%).

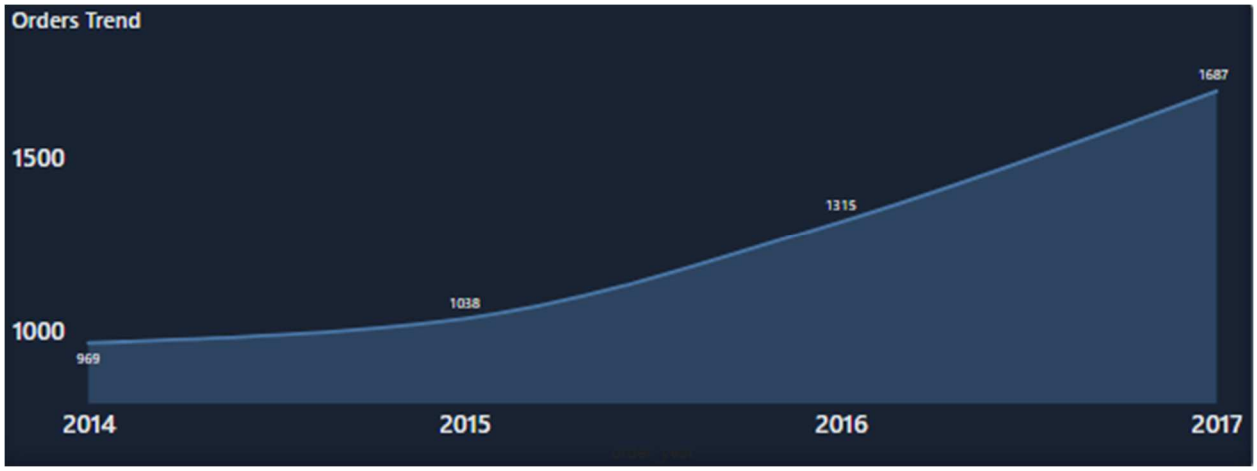
The Consumer segment accounts for nearly half of total orders, making it most active customer group. other segment contributes significantly less in comparison.

Suggestions:

- Focus marketing and retention strategies on Consumer customers.
- Provide offers or loyalty programs to maintain high order volume.
- Improve engagement in Corporate and Home Office segments to increase orders.
- Review strategy of Corporate and Home Office segment.

22. Is the number of orders increasing over time?

Ans:



Yes, the number of orders increase over time. Orders increase from 969 in 2014 in 2014 to 1038 in 2015, then to 1315 in 2016, and reached to 1687 in 2017.

The order trend shows a consistent upward pattern, with a noticeable increase especially after 2014. The growth between 2016 and 2017 is particularly strong.

The business is experiencing steady growth in terms of order volume, indicating increasing demand over the years.

Higher number of orders can lead to increased sales, but it is important to ensure that this growth also translates into profit.

Overall, the increasing order trend reflects positive business growth, but it should be aligned with profitability to ensure sustainable performance.

23. What is the average shipping time between order date and ship date?

Ans:

The average shipping time between order date and ship date is approximately 3.96 days**.**

This means that, on average, it takes around 4 days for an order to be shipped after it is placed.

The shipping process is relatively consistent and falls within a reasonable time range for order fulfillment.

A shipping time of around 4 days indicates decent operational efficiency, but there may still be opportunities to reduce delivery time and improve customer satisfaction.

Overall, the average shipping time is acceptable, but optimizing logistics and reducing delays can further enhance performance.

24. Which shipping mode is used the most?

Ans:

The most used shipping mode is Standard Class, with 2994 orders, followed by Second Class (964 orders), First Class (787 orders), and Same Day (264 orders).

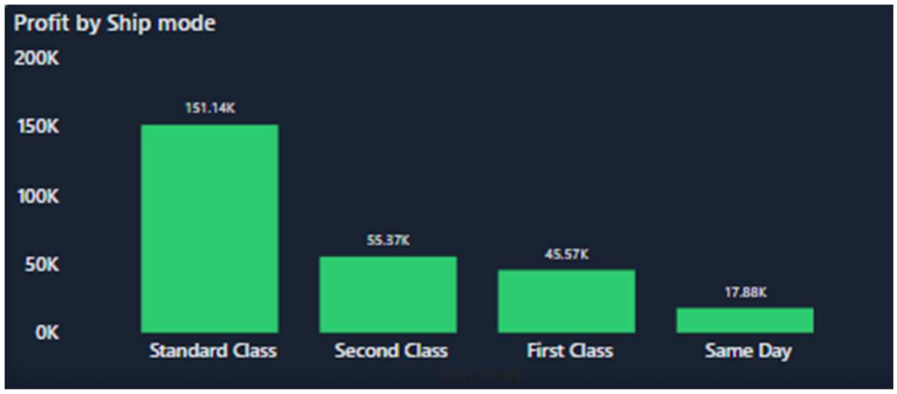
Standard Class is used significantly more than other shipping modes, indicating that most customers prefer a cost-effective delivery option over faster shipping.

There is a strong reliance on Standard Class shipping, which suggests that customers prioritize lower shipping cost rather than speed.

Standard Class is the dominant shipping mode, but balancing cost and delivery speed could improve overall customer experience.

25. Which shipping mode generates the most profit?

Ans:



The Standard Class shipping mode generates the most profit, with approximately $151.14K, followed by Second Class ($55.37K), First Class ($45.57K), and Same Day ($17.88K).

Standard Class contributes significantly more profit than other shipping modes. This is likely because it is used for the majority of orders.

There is a strong relationship between usage and profit since Standard Class is the most used shipping mode, it also generates the highest profit.

Relying heavily on Standard Class helps in maintaining profitability, but faster shipping options are underutilized and contribute less to profit.

Standard Class is the most profitable shipping mode, mainly due to its high usage. However, there may be opportunities to improve profitability in other shipping modes.

Conclusion

From the analysis, it is clear that the business is growing in terms of sales and number of orders over the years. However, this growth is not fully reflected in profit, as profit margin has slightly decreased and losses have also increased.

The main issues identified are high discounts, especially above 50%, and loss-making sub-categories like Tables, Bookcases, and Supplies. Some categories like Furniture generate good sales but contribute very little to profit, while Technology and Copiers perform well in both sales and profitability.

In terms of operations, the average shipping time is around 4 days, which is acceptable, but there is still scope for improvement. Standard Class is the most commonly used and most profitable shipping mode.

Overall, the business should focus more on improving profitability rather than just increasing sales by optimizing pricing strategies, controlling discounts, and reducing losses from underperforming products.